

ADTA 5900/5750.501: Applied Natural Language Processing

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Assignment 5

1. Overview

The rise of cloud computing has been a facilitator of the emergence of big data. Cloud computing is the commodification of computing time and data storage using standardized technologies.

Big data is a term to describe large volumes of data that can be both structured and unstructured. These enormous volumes of data overwhelm the digital world every second. However, it is not the amount of data that is important. It is what we can do with the data that matters: Big data analytics can provide insights that lead to better decisions and strategic moves.

It was the emergence of cloud computing that made it easier to provide the best of technology in the most cost-effective packages. Cloud computing has not only reduced costs but also made a wide array of applications available to companies of all sizes: small, mid-sized, big, and giant corporations.

2. Google Cloud Platform (GCP): Service Account

Various enterprise services of Google Cloud Platform (GCP) will be used in classwork. When using cloud services to run applications, cyber-security is one of the top-priority concerns. In order to use the services to create and run Natural Language Processing (NLP) applications in GCP, the user is required to set up a service account that can be used in the GCP's sophisticated authentication system.

IMPORTANT NOTES:

--> All the documents posted on the Canvas page **GOOGLE CLOUD PLATFORM: GCP for Deep Learning – TF2** should be used for HW 1.

--> All the documents posted on the Canvas page **GOOGLE CLOUD PLATFORM: GCP for Natural Language Processing (NLP)** should be used for HW 1.

--> All the **class assignments**, including the **midterm** and the **final project**, should be done using the services of GCP.

3. Homework 5: Assignment Format

Homework 4 is assigned as a team assignment. It means that all the student members of a group will collaborate with each other while working on the assignment.

However, each student is required to submit his/her report individually. In other words, students work on the assignment as a team but submit the work as if he/she worked on the assignment by himself/herself.

4. Homework 4: General Assignments

Student members of a group are assigned to create a conversational AI system for a small business organization (any kind of business) similar to that of the retailer G-Records that has been done in Homework 3 and the midterm.

5. PART I: Conversational AI System: System Analysis (100 Points)

The student is required to submit a project analysis report as the solution to Homework 5.

The report must include the following sections:

1. Introduction

- a. Provide an overview of the project.

2. Problem statement

- a. Discuss in detail what problem (business, technical, ...) the student is trying to solve with this project.

3. System Requirements Analysis (HW 4)

- a. Business requirements
 - i. Discuss in detail the business requirements of the conversational AI system.
- b. Technical requirements
 - i. Discuss in detail the technical requirements of the conversational AI system.
- c. Data requirements
 - i. Discuss in detail the data requirements of the conversational AI system .

4. Feasibility Analysis

- a. Technical feasibility analysis:
 - i. Can we complete the project successfully as required?
 - ii. Discuss any technical risks while working on the project
- b. Business feasibility analysis:
 - i. Will the project provide good business values after its completion?

- ii. Discuss any financial risks while working on the project, e.g., running out of funding
- c. Operation feasibility analysis
 - i. If we build the system, will it be used by the organization as expected?
 - ii. Discuss any risks that may hinder the deployment of the system after its completion.

5. Project Management

- a. Discuss in detail the timeline of the project
 - i. What kinds of major tasks to be done?
 - ii. What major phases need to be done until the completion?
 - 1. Due date of each phase or task.
- b. Discuss in detail the human resources needed for the project
 - i. How many people, in total, are assigned to work on the project?
 - ii. How many people are assigned to work on each major phase of the project?
 - iii. For each phase, who do what?

6. Conclusion

- a. Provide a short paragraph to express opinions about how the project will be done to conclude the report.
 - i. For example: There are many challenges but will be successful with great efforts.

6. HOWTO Submit

The student is required to submit all the sections, i.e., submission requirements, in a Microsoft Word document that is sent to the instructor (Thuan.Nguyen@unt.edu) as an attachment to a UNT email.

The subject of the email must be:

- “[ADTA 5750: Assignment 5 – Submission.](#)”

Due date & time: 11:00 PM – Wednesday 11/20/2024